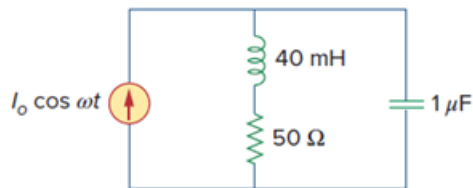


Problem

Use magnitude and frequency scaling on the circuit of Fig. 14.79 to obtain an equivalent circuit in which the inductor and capacitor have magnitude 1 H and 1 F respectively

Figure 14.79:



Step-by-step solution

Step 1 of 1

$$R = 50\Omega, \quad L = 40\text{ mH}, \quad C = 1\mu\text{F}$$

$$L' = \frac{K_m}{K_f} L \rightarrow 1 = \frac{K_m}{K_f} (40 \times 10^{-3})$$

$$25K_f = K_m \quad -(1)$$

$$C' = \frac{C}{K_m K_f} \rightarrow 1 = \frac{10^{-6}}{K_m K_f}$$

$$10^{-6} K_f = 1 |_{K_m} \quad -(2)$$

Substituting (1) in to (2)

$$10^{-6} K_f = 1 |_{25K_f}$$

$$K_f = 0.2 \times 10^{-3}$$

$$K_m = 25K_f = 5 \times 10^{-3}$$